The University of Texas at Austin
Texas McCombs
MS Business Analytics
McCombs School of Business

BUSINESS ANALYTICS CAPSTONE
Project Description

1. **Motivation(s):** Please provide a brief background on your business and articulate which business opportunity or problem motivates this data analytics project. Please also briefly articulate how this data analytical project is likely to add business value to your firm.

Business background
Zebra Technologies (NASDAQ: ZBRA) is an innovator at the edge of the enterprise with solutions and partners that enable businesses to gain a performance edge. Zebra's products, software, services, analytics and solutions are used to intelligently connect people, assets and data to help our customers in a number of industries make business-critical decisions. These industries include: Retail and E-commerce, Manufacturing, Transportation & Logistics, Banking, Healthcare, Public Sector and Hospitality.

With a rich history of innovation, Zebra is recognized as an industry leader in Enterprise Mobile Computing, Barcode Scanning, Specialty Printing, RFID Reader and RFID Printing, Workforce Management and Magic quadrant for Indoor Location Services.

Zebra empowers organizations to thrive in the on-demand economy by making every front-line worker and asset at the edge visible, connected and fully optimized. Supply chains are more dynamic, customers and patients are better served, and workers are more engaged when they utilize Zebra innovations that help them sense, analyze and act in real time.

Business opportunity or problem that motivates this project
Zebra operates a network of over 50 Service locations spanning the globe. These locations perform repairs and triage on customer devices. To effectively meet customer obligations on turn-around time and quality, Zebra service planners must plan the demand and supply to ensure sufficient spare parts inventory is maintained at each location.

Through this project we would like to identify strategies to improve overall inventory performance, to achieve the best possible material fill rate with the lowest total inventory investment. Influencing factors in determining target inventory levels may include the following

- Variability in material usage
- Forecast accuracy
- Supply variability, including repair/reverse logistics
- Transportation times
- Inventory segmentation
- Inventory pooling strategies (by or within region)
- Target material fill rate
- Import/export regulations and ease of customs, influencing transportation times
- Duty and Tariffs

Students are encouraged to suggest additional factors to be incorporated into the models.

If done well, how and why is this project likely to add business value to your firm?
Optimized inventory plans will allow the effective service of repair contracts with the lowest practical inventory level across the network. This can have a significant impact on balance sheet performance (inventory), customer satisfaction and P&L performance through reduced Excess and Obsolete (E&O), transportation and other costs.

2. **Proposed business question(s):** Please frame the business question(s) that you are asking the student team to answer in this project with data analytics.

List of business questions to be answered in this project

Question 1: What is the optimal inventory level for each part/location combination within the service organization, including strategic network design recommendations

Question 2: Can we improve or maintain our customer satisfaction scores/material fill rate with a lower network wide inventory level?

Question 3: Are there critical limitations to address prior to operating under the optimal inventory levels (e.g. supplier delivery performance impacting lead times and tariffs)?

3. Data: please see the data related question in the bullet points below and address them to the best of your knowledge.

DATA:

- Sponsor must assess the variables captured in the data that will be provided to students and include a statement about the feasibility to answering the business questions posed with the available data. For example, if the question posed assumes that multiple data sets will be matched and analyzed, is matching feasible at the level of analysis required?

  - Students will be provided with access to data needed to develop the analysis, variables include forecast data, product family data, product part numbers/SKU's, etc. Most data utilized for this project will be structured, but some unstructured data may also be made available to students throughout the course of the project. Examples of data sources include Zebra Data Lake (Google Cloud Platform), Oracle ERP data, Supply Chain data, Agile PLM system data, static data (Excel), etc.

- A list of available variables (not data per se, just variable names) so that instructors and students can also form an opinion about the available data and whether it would allow the students to answer the business questions.

  - Below are examples of the data sources we will provide for analysis

    - Item Master data (part number, price, suppliers and any additional data sets within master data)
    - Historical parts usage in repair depots, presented in monthly buckets
    - Current and historical forecasts
    - Order data (shipments from DC to depot) to assess transportation time and variability
    - Supplier PO and delivery data
    - Repair volumes by product
    - Service BOM's

- The size of the data that will be shared with students. If the data is too large to share in its entirety, a representative sample could be shared with students.

  - Data needed for this project will be made available to the students via secure network connection to Zebra's data lake using a Zebra-supplied computer. Students will develop all analytics entirely within the Zebra IT domain. At the conclusion of the project/capstone, Zebra computers will be returned to Zebra & analytics IP will be retained by Zebra.

- IT to provide estimate of size of data

- Sponsors' own assessment of data quality; and statement of data preparation work that students will be expected to do.

  - Data quality is considered high and reliable, with some known inconsistencies where assumptions may need to be made

- If sponsors expect students to collect additional data over and above what the sponsor will provide, the nature of the additional data and the data collection methods ought to be stated in the proposal. The sponsors should also plan for financial resources that might be required to collect the additional data.

  - Additional external data collection not anticipated but will be considered should a case be presented by the students. Zebra will consider funding after evaluation of business case

- Sponsor must disclose if the available data is in structured or unstructured form. Structured

  - The data provided will be structured

- If the success of the project relies heavily on the use of unstructured data, it must be disclosed in the project proposals

  - N/A

4. **Data processing environment:** please state if students will be expected to work on the data in your company's IT environment; or if you will transfer the data to students. Please also assess whether the access or data transfer can be secured within the deadlines stated in Exhibit 1.

Students will be required to work on the data wholly within Zebra's IT environment; no Zebra data will be exported outside of the Zebra domain. Provisions have already been made to provide system/data/tool access to students, along with loaner laptop computers for the duration of the capstone project. Zebra's standard analytics tool suite includes GCP, Databricks, Big Query, Python.

5. **NDA:** please state if students will be asked to sign an NDA to gain access to the data set.

Yes

6. **Key business and IT contacts:**
   We recommend that the sponsor identify two main contacts to the student team: (a) business contact who understands the business domain of the project and can provide inputs on business related questions and decisions; (b) IT contact who can address data access related questions of the student team.

   Contact persons named should review the project timetable in Exhibit-1 and make a commitment to be available to meet the deadlines.

   a.

Business domain contact (please provide name and contact information)
Chris Brown: chris.brown@zebra.com / +1 631 820 1045

   b.

IT/Dataset support contact (please provide name and contact information)
Leticia Cruz: leticia.cruz@zebra.com / +1 956 496 4747

7. **Project delivery expectations: (i.e. format, location, visualization requirements, desired modeling tool, etc.)**

Output for this project shall be delivered in visualizations that empower business users with limited data science background to intuitively operate, including conducting “what-if” analyses. Visualizations are to be delivered in PowerBI; however, Zebra is open to discussion of potential suggested alternatives, should the team wish to make such recommendations

8. **Expectations about the composition of student team:**
   a. **Student backgrounds:** If the proposed project has any particular background and skills requirements (e.g., business, engineering, math, humanities / US citizenship versus International, etc.), please list them here. While it may not be feasible to meet all requirements listed here, if there is room, we will try to accommodate the requirements in matching students to projects.

Basic understanding of inventory management, BOM’s, Supply Chain & repair and general Business knowledge would be helpful. Supply Chain Planning, including repair demand forecasting and understanding of reverse logistics concepts would be ideal but not essential for this project. Having prior knowledge in these specific areas is beneficial but not essential. Students will have the opportunity to learn about these topics throughout the project.

b. **Are you open to working with a student team from the Working Professionals MSBA program?** Working Professional students balance their coursework with full-time employment. The capstone schedule is identical to the full-time MSBA program, and their course curriculum is the same.

No

c. **Are you open to participation from senior undergraduate students in the Business Analytics (BAX) program?** The MSBA students will always be the primary team. The level of the undergraduates' involvement is under your discretion, and can range from working in parallel with the MSBA team, collaborating with them, or shadowing them throughout the capstone. Projects using only public data or without strict NDA requirements are often best.

No